Engineering Change Notice(ECN)

ECN-D-2609-04

(ECP AP EFEM FRAME Part Change)

FA Research ME 2 Team

Written	Review	Confirm
곽수민 프로	최동수 팀장	이태경 본부장
09/02	09/02	09/02

NINEBELL

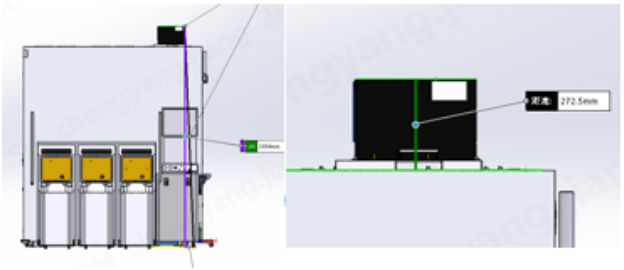
1. Engineering Change Request

NINEBELL

설계 변경 요청서(ECR)
(Engineering Change Request)

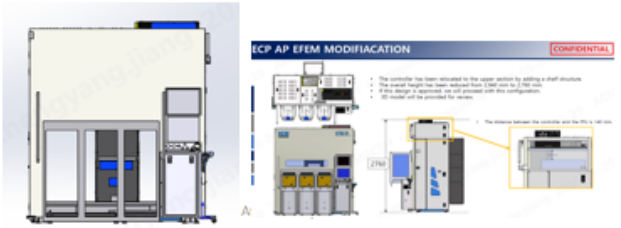
				Customer	ACMK
				ECR No.	
Requesting department	Name	CHENGYANGJIANG	Change ITEM	☑ Customer Requst ☐ Cost Reduction ☐ Quality Improve ☐ Work Efficiency ☐ ETC	
	Date	2026.6.30			
Request project	Category	Project Desc.	Type	변경 Level (Change Level)	☐ Level 1 — Design structure changes are required
	☐ MP ☐ ES	ECP ap 337	ap		☐ Level 2 — Simple part and work method change

Before(Problems and Change Requests)



Due to the height limitations of the customer's fab workshop, the current EFEM of the required height cannot meet the usage needs. Therefore, we need you to adjust the height of the EFEM.

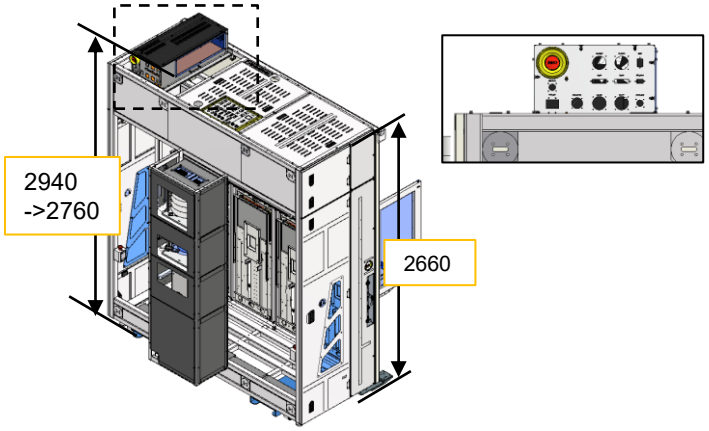
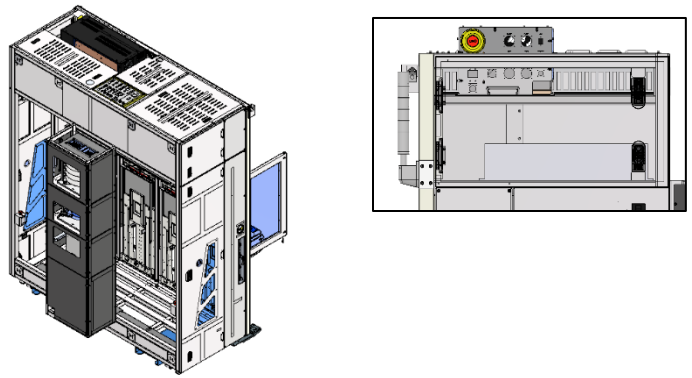
After(Expected Effect)



Adjust the overall height of the EFEM

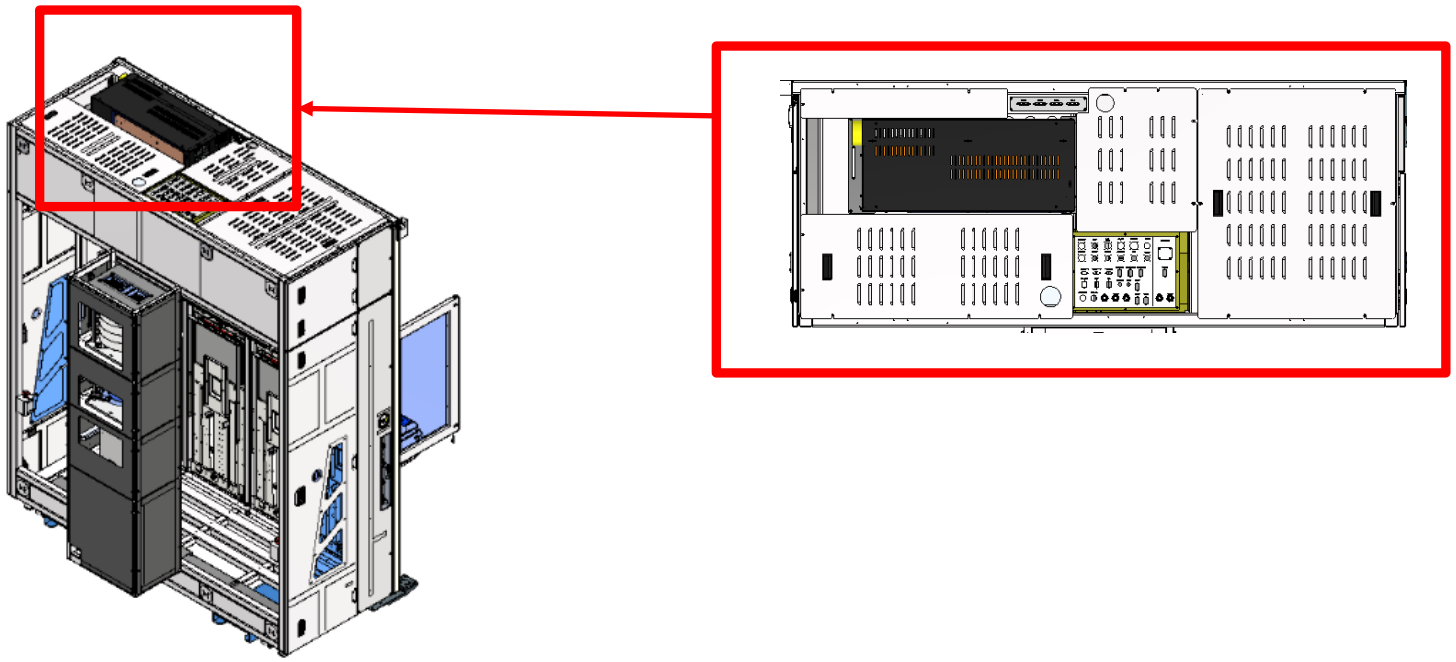
Related Division	☑ Mechanical design	Potential Issue	☐ Instrument & tool additional cost incurred
	☐ Electrical design		☐ Work process change and work time
	☑ Manufacturing		☑ Mechanical failure
	☐ Purchasing & Materials		☐ ETC
	☐ Quality		

2. Engineering Change Notice

적용 장비 타입 Applicable project type	ECP AP EFEM	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) □ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) □ 생산성 향상 (Increase productivity) ■ 부품 단종 (Part discontinuation) □ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	별도 기입 필요 X
변경 상세 사유 Detailed reason for change	일부 ACM 공장은 천장 높이가 3m로 제한되어 있어, EFEM 설치 시 ACM 천장과의 간섭이 발생할 수 있습니다. 이에 따라 컨트롤러를 프레임 내부로 배치하여 EFEM의 전체 높이를 낮추었습니다. As some ACM factories have a limited ceiling height of 3m, interference with the ACM ceiling may occur when installing the EFEM. Therefore, the controller was relocated inside the frame to reduce the overall height of the EFEM.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail		➔ 프레임 안쪽에 Control box 선반 용접 ➔ 전체 높이 변경 (2940->2760mm) ➔ Welding of the Control Box shelf inside the frame ➔ Overall height changed (2,940 → 2,760 mm)	

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3. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표기 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / N	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approval	Y / N
승인 날짜 Approval date	1. Ninebell : 2026.09.08 2. Customer : / /		나인벨 승인 Ninebell approval	QC 2026.09.08 PASS	고객사 승인 Customer approval	ACM confirms chengyang jiang 9/11/2026
적용 일자 Effective date	2026.07.23(E-BOM)		최초 적용 장비 Initially Applied Equipment	ECP337~		

4. Attachments

Part No.		
Before	→	After
EE55-F1000-6	→	EE55-F1000-8
EEAP-F20P9-0	→	EEAP-F20P19-0
EEAP-F20P8-0	→	EEAP-F20P20-0
		EEAP-F20P21-0

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